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Spanish diminutives in mother–child conversations*

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ABSTRACT

The present study examined gender and age patterns of diminutive use in conversations between 32 Spanish-speaking Peruvian mothers and their three- and five-year-old children. Results confirm previous findings concerning both parents' greater use of diminutives with younger children and children's early acquisition of this complex aspect of morphology. However, findings do not support previous studies on gender differences in parental use of diminutives with young children. Results also revealed that mothers' and children's imitations of their interlocutors' diminutized words promoted their interlocutors' overall diminutive use. This finding highlights the acute sensitivity of both speakers to each others' language and the potential role of imitation in older children's language development.

INTRODUCTION

Diminutives are a near-universal component of human speech (Jurafsky, 1996; Bauer, 1997). Generally this class of derivational morphology is characterized by a literal meaning of smallness, as well as a variety of positive hypocoristic meanings such as endearment or tenderness (Sifianou, 1992; Dressler & Barbaresi, 1994; Bulatetska, 1996). As they carry a wide range of meanings and emotional overtones, diminutives, together with augmentatives, are also known as evaluative or expressive morphology.

Linguists have long been intrigued by the syntactic, semantic, and pragmatic complexity of diminutives. Dressler & Barbaresi (1994) date examinations of

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diminutive use as far back as Greek and Roman times, when semantic and pragmatic aspects of diminutives were of primary interest. The debate concerning the meanings embedded in diminutive use is still current, with some linguists arguing that diminutives are predominantly markers of subjectivity and emotionality, such as affective proximity and detachment (Alonso, 1961; Inchaurralde, 1997), and others that meanings vary from denotational smallness to a range of connotative references such as playfulness or intimacy (Ettinger, 1974, cited in Dressler & Barbaresi, 1994; Jurafsky, 1996). Linguists' interest in diminutives has also focused on their structural characteristics, including the rules for attachment to stem words, and the word-class of the landing sites of diminutive forms, in particular, the potential existence and nature of universal hierarchies. Such cross-linguistic or universal hierarchies suggest, for instance, that 'for a word-class to be used as the base in evaluative morphology in a particular language, word-classes from each step above that [word-class] must also be so used in that language' (Bauer, 1997: 540). Bauer (1997), as an example, proposes the following hierarchy: (a) Noun, (b) Adjective/Verb, (c) Adverb/Numeral/Pronoun/Interjection, (d) Determiner.[1] Other predictions concerning these universal hierarchies allude to diminutives' semantic and pragmatic features. These predictions stipulate, for instance, that the lower down a word-class is on the hierarchy, the less productive and 'the greater the importance of subjective' force of the diminutive in that class (Nieuwenhuis, 1985: 221, cited in Bauer, 1997: 540). Much of the work in this area is limited, however, in that it relies on researchers' intuitions concerning speakers' pragmatic and semantic intent.

Other lines of inquiry with a stronger empirical base have found diminutives to be strongly associated with female speech (Daltas, 1985; Andrews, 1999), and with speech directed to or about young children (Ferguson, 1977). Yet despite the apparent saliency of diminutives in child-directed speech, few studies have examined the mechanisms which support young children's acquisition and use of diminutives, and even fewer have analysed how this development varies by age or gender. This imbalance is in part due to the tendency of child language researchers to focus on the acquisition of open-class vocabulary and word order, rather than closed-class morphemes (Peters, 1995). This imbalance also results from the fact that much of the early research on language acquisition was conducted on native speakers of English, a language which has a relatively impoverished and unproductive diminutive system (Peters, 1995). English relies primarily on *-y/ie*, as in *doggie* or *piggy*

[1] Bauer notes that her hierarchy does not imply that both verbs and adjectives must be diminutizable for nouns to be diminutizable; either verbs or adjectives alone is sufficient. This presumably holds true for level (c) as well, that is diminutization of adverbs, numerals, pronouns or interjections would be sufficient to allow for adjectives or verbs to take the diminutive affix.

(Berko Gleason, Perlmann, Ely & Evans, 1994). Sifianou, for example, observes that while diminutives in Greek and English express familiarity, informality, and endearment, English diminutives are far less flexible and less productive in conveying emotions or attitudes, largely because 'English words accepting diminutive use are limited, and have more or less fixed meanings' (Sifianou, 1992: 157).

Diminutives in children's and parents' language

Not surprisingly, most studies investigating the developmental trajectory of diminutives have targeted speakers of European languages, which tend to make frequent use of evaluative morphology. Many of these studies observe that despite their semantic, structural, and pragmatic complexity (Dressler & Barbaresi, 1994), diminutives are one of the earliest, if not the earliest morphosyntactic form to be acquired, with both simplex and diminutized forms appearing in the speech of very young children (e.g. Ferguson, 1977; Ceccherini, Bonifacio & Zocconi, 1997).

Systematic investigations of the frequency of use of diminutives among young children and their mothers likewise suggest that children attain adultlike usage patterns at relatively early ages. Stephany (1997), for instance, in her naturalistic study of diminutives in the speech of two monolingual Greek children, found that at 1;9 both children used the three most common diminutive suffixes with the same relative frequency as their mothers. Similarly, Gillis (1997), in his longitudinal study of the acquisition of Dutch diminutives in the natural speech of one child between the ages of 1;5 and 2;5, reports that by early in the second year of life, the patterns of diminutive use, as measured by the percentage of diminutive word forms, were comparable to that of the mother. Likewise, in her longitudinal diary study of the development of words for 'mama' and 'papa' among two Latvian-speakers, Rūķe-Draviņa (1976) reports that the ability to use diminutives productively appears around two years of age. The apparent early acquisition of evaluative morphology has led researchers to speculate about the facilitative role that diminutives might play in language acquisition (Ferguson, 1977). For instance, Olmsted has argued that because Russian diminutives are characterized by completely predictable declensional endings, no third declension, and limited stem alterations, they provide a simpler subset of morphemes which children can master with relative ease prior to 'venturing into the wilder fuller nominal system of Russian at large' (Olmsted, 1994: 166).

Of course the fact that children use diminutives with similar frequencies as their interlocutors does not mean they have mastered all aspects of the diminutive system; correspondingly, there is evidence that the developmental trajectory, especially that for semantic and pragmatic functions, extends

much later and is more complicated than suggested by the studies above. Much of this evidence comes from experimental data, such as Berko's (1958) 'wug' study, in which she reported that none of the 51 children (aged 5;6 to 7;0) was able to use the diminutive suffixes which were employed by the study's adult participants (such as *wuglet*, *wuggie*, *wugette*, and *wugling*). Likewise, Bates & Rankin (1979), in their second of two studies on diminutive acquisition, performed comprehension experiments with 84 native-Italian-speaking children ranged evenly from 2;0 to 6;0. They reported that children's level of performance differed as a function of age and the semantic nature of particular morphemes tested. The authors concluded that the development of evaluative inflections was not complete by the second year of life as past research suggested, and that acquisition patterns depended on the connotative complexity of individual morphemes. Similarly, Snow, Smith & Hoefnagel-Höhle (1980) experimentally examined the acquisition of the morphological rules for plural, agentive, and diminutive suffixes among two groups of native Dutch speakers (seven- and twelve-year-olds) and three groups of Dutch second language learners (five to ten, twelve to eighteen, and adults). Of interest are the results for the 16 native Dutch children who were tested on the application of diminutive formation rules: whereas only 87% of seven-year-olds could apply all of the rules of the full diminutive suffix system, 100% of the twelve-year-olds could do so. The authors noted that the rules for diminutive formation seem to be guided by lexical acquisition and only fully acquired 'after a certain number of lexical items which demonstrate (a specific) rule have been learned both in their standard and their diminutive forms' (Snow *et al.*, 1980: 550). Lastly, Herrera & Carvallo's (1987) analysis of diminutives in the speech of 12 Spanish-speaking children (seven- and ten-year-olds) in Caracas, Venezuela, suggests that children of both ages and genders have mastered both the denotational meaning of smallness and the connotational meanings of affection, but have yet to master common pragmatic functions found in adult language, such as expression of politeness or disparagement.

These differential findings concerning when diminutives are acquired largely reflect the varied interpretations of ACQUISITION (e.g. production of both simplex and diminutive forms vs. usage patterns which mirror the frequencies of parents vs. adultlike range of pragmatic functions). Overall, the limited research in this area suggests that children are capable of producing diminutive forms quite early in natural conversation, although full acquisition of the structural, semantic, and pragmatic system possibly continues through middle childhood (ten to twelve years of age). Many aspects of this trajectory, however, are less than clear. In particular, while studies suggest that children attain adultlike usage patterns (in terms of the percentage of diminutized words) at around two years of age, it is not known to what extent these patterns truly reflect adult usage norms. For instance, younger

children – more than older children or adults – might rely on imitation of their interlocutors' diminutives, rather than independently introducing their own diminutized forms. Equally unclear from the above studies is how caretakers' use of diminutives supports children's use and acquisition; for instance, do mothers of younger children repeat or imitate diminutive forms more than mothers of older children, and more broadly, what is the overall relationship between interlocutors' usage modes? Such questions – which are addressed in the present study – are critical to understanding the nature of the developmental trajectory for diminutives, but also potentially provide insights concerning how interactional processes such as imitation and repetition might support children's morphological development more generally.

While many aspects concerning how diminutives are used and acquired are far from clear, so are issues surrounding how this trajectory might vary across genders as a result of the varied input which boys and girls seem to receive. The limited research in this area suggests that boys and girls hear different types and amounts of diminutized forms. For instance, Berko Gleason *et al.* (1994), in one of the few studies of diminutives in English, examined how 88 parents use diminutives with their small children (1;2 to 5;0). They found that at 1;2, mothers produced roughly the same number of diminutives to boys as to girls. However, by two years of age, parents began to use more diminutives with girls than with boys. Since use of diminutives is frequently associated with speech to younger children, as might be expected, the total number of diminutives and the range of diminutive types used by parents decreased as the children grew older. However, parents of girls tended to employ a wider range of diminutized words, whereas parents of boys used a more conventional or frozen set. Yet as this is the only study which empirically investigates potential gender differences in parental use of diminutives and as English makes limited use of diminutized forms, it is unknown whether such differences are language-specific.

Spanish diminutives

In sharp contrast to English, Spanish has a rich and complex evaluative morphology system (Alonso, 1961; Inchaurralde, 1997); yet relatively little empirical research has been conducted with actual speakers, and none among children under seven. This is surprising given that the diminutive has long been a prominent feature of the Spanish language. The relative importance of its everyday usage, however, has changed over the centuries. During the Middle Ages, 'the common forms were *-illo*, *-illa*, *-uelo*, *-uela*, *-iello*; but there were new affixes that appeared towards the end of the Middle Ages (*-ico*, *-ito*) which began to compete with them' (Inchaurralde, 1997: 138). Presently, there is wide variation in preferences towards certain diminutive forms both across different regions of Spain and within the Americas; yet by

far the most popular in both regions is *-ito/-ita*. Indeed, this affix 'may well be considered one of the most productive morphological processes of Spanish' (Jaeggli, 1980: 142). Zuluaga (1991) reports, for instance, that of the 175 diminutives found in the novel, *Cien Años de Soledad*, '*One Hundred Years of Solitude*' (García Márquez, 1967), 142 (81%) are formed with *-ito*. More pertinent to the present study, Herrera & Carvallo (1987) find that of the 822 diminutives in their sample of children's speech, 769 (93·5%) are formed with *-ito*.

Although *-ito* is generally considered to constitute one diminutive affix, in fact it is formed through the insertion of three distinct infixes: (a) *-it-* (as in *tonto/tontito*); (b) *-sit-* or *-cit-* (as in *corazón/corazonsito* or *corazón/corazoncito* 'heart'); and (c) *-esit-* or *-ecit-* (as in *cruz/crucesita* or *cruz/crucecita* 'cross') (Jaeggli, 1980; Ambadiang, 1997). (This spelling difference is a reflection of the non-standardized and informal nature of diminutives, not unlike *doggie/doggy* in English.) These affixes are not found in free variation; rather, their use is governed by rules relating to the gender, number of syllables, and segmental form of the final syllable of the base form.

As in many other languages, diminutives in Spanish are not only structurally but also semantically and pragmatically complex. As an example, the same morpheme attached to the same lexical base potentially expresses quite different meanings depending on the norms of the particular speech community. For instance, in Mexican Spanish, the diminutized form *ahorita* carries an intensifying force; in Dominican Spanish, in contrast, it has an attenuating effect (Jurafsky, 1996: 534): (1) Mexican Spanish *ahorita* (NOW-DIM/FEM)[2] 'immediately, right now', (2) Dominican Spanish *ahorita* (NOW-DIM/FEM) 'soon, in a little while'. Thus, Spanish evaluative morphology is complex and shares much in common with the systems of other languages, but has received little attention in terms of its acquisition and use patterns.

As is clear from the brief review above, several key areas concerning diminutive use and development remain unaddressed. First, there are no studies on the development of Spanish evaluative morphology with children under the age of seven, and second, there are few data on the development of diminutive forms among any children between three and seven years of age, a potentially critical period for morphological development (Peters, 1995). Third, whereas the analysis of Berko Gleason *et al.* (1994) suggests that the types and amount of diminutized forms to which children are exposed varies by gender, we have no systematic studies of how this input influences girls'

[2] All nouns in Spanish are either masculine (typically words ending in *o*) or feminine (typically words ending in *a*). The diminutive affix reflects the gender of the base form, sometimes marking the gender of the word more overtly; for instance, *cruz*, which is not obviously masculine or feminine, becomes *crucecita*, which is clearly feminine.

and boys' varied acquisition and use of diminutized forms. Fourth and most significant for the purposes of the present study is the lack of systematic research concerning how diminutives are used in everyday conversations between children and their mothers; what relationship exists between mother and child usage patterns; and how these interactions potentially facilitate children's diminutive use and development. Also notable from the above review is the tendency for developmental studies of diminutives to rely on data from one or two children, or to elicit data through experimental methods. Given the highly varied and individualized nature of diminutive use patterns (Andrews, 1999) and their association with informal, intimate, and emotional communication, these methods are less than ideal (*cf.* Derwing, 1976).

Given the above gaps in the literature, this paper has three aims: (a) to provide basic descriptive data concerning the use of Spanish diminutives in mother–child conversations, (b) to explore age and gender variations in mothers' and children's diminutive usage, and (c) to analyse the relationships between mothers' and children's patterns of diminutive use. In addition, this study seeks to overcome methodological obstacles outlined above through a quantitative analysis of naturalistic conversations between a relatively large sample of Spanish-speaking mothers and their children.

METHOD

Participants

Thirty-two mothers and their preschoolers participated in this study. All families resided in Lima, the capital of Peru. Lima is a modern city with a population of approximately 7·5 million people, located on the Pacific coast of South America. Families were recruited through institutional and personal contacts of the first author. Mothers' ages ranged from 24;0 to 39;0, with a mean age of 31 years. All families spoke only Spanish, specifically Limeño[3] Spanish. All parents were born and raised in Peru and the majority of parents (81%) were born and raised in Lima.

By Peruvian standards, families' socioeconomic status was middle to upper-middle class. All parents had completed high school (11 years in the Peruvian system) and attended post-high-school institutions. All fathers and 26 mothers (81%) had attended universities; the remaining six mothers attended institutes of higher education (e.g. schools for interior design). Eighty-eight percent of the sample (28 families) were members of two-parent homes; three mothers were separated; one was divorced. All fathers and

[3] Limeño literally means pertaining to Lima; it is used to refer to the people of Lima, their language variety, and their customs. For a review of Peruvian Spanish, see Lipski (1994).

TABLE 1. *Sociodemographic profiles of participants by age group*

	Three-year-olds ($N=16$) M (range)	Five-year-olds ($N=16$) M (range)
Children		
Age	3;7 (3;0–4;0)	5;5 (5;0–5;11)
Age of first words	10 mo (5–18)	10 mo (6–20)
Birth order		
First-born	10	12
Second-born	2	1
Later-born	4	3
Mothers		
Age	29 (24–35)	32 (27–39)
Years of education	16 (12–18)	17 (13–20)
N employed	11	13
Fathers		
Age	35 (28–41)	35 (29–40)
Years of education	16 (14–24)	17 (12–23)
N living with family	15	13
Home		
Household size	7 (3–11)	6 (4–8)
N of children	2 (1–4)	2 (1–3)

24 mothers (75%) worked outside the home. All fathers held professional positions (e.g. engineers, university professors, entrepreneurs, veterinarians, managers, politicians). Mothers held semi-professional and clerical positions. Households were composed of both nuclear and extended family members. There was an average of 6 to 7 members per household, which could include grandparents, aunts, uncles, cousins and live-in household help. Families had an average of 2 children (range 1–4 children per family).

Children were of two age groups: three-year-olds ($M=3;7$, S.D. $=3{\cdot}8$ months) and five-year-olds ($M=5;5$, S.D. $=3{\cdot}4$ months), each divided equally by gender. According to parental reports, children were developing normally. The mean age reported for children's first words was 10 months. All but one child attended schools or day care centres. Twenty-two children (68·7%) were first-born; three (9·4%) were second-born; and seven (21·9%) were the youngest in their family (see Table 1 for sociodemographic profiles by age group).

Procedure

A Spanish-speaking Peruvian researcher visited mothers and children in their homes once; visits lasted approximately four hours. Each mother was asked to engage in a conversation with her child about six events the child had experienced recently. Mothers were instructed to talk about one specific

event at a time and to avoid retelling movies or storybooks. Written instructions were also provided, but no further specifications concerning the task were given.

The researcher explained to children that she was interested in tape-recording their voices and that they could hear themselves at the end of the conversation. Tapes were allowed to run until mothers and children were finished; the average time spent on the task was approximately 45–60 minutes. The researcher was in the home at the time of the recordings, but neither she nor anyone else was present in the room where the recordings of mother–child conversations took place.

Coding categories

Audiotapes were transcribed by a native Spanish speaker following a standardized format, Codes for the Human Analysis of Language (CHAT), available through the Child Language Data Exchange System (CHILDES; MacWhinney, 2000). Transcriptions were verified by the first author, a native of Lima. Conversations were segmented into two types of speech events: (a) narrative exchanges identified using verbal and nonverbal cues provided by mothers and children (e.g. comments such as: 'now let's talk about ...,' 'tell me about the time ...,' 'I don't wanna talk anymore,' 'is there anything else you want to tell me?'); and (b) spontaneous conversations between narrative exchanges (e.g. deciding what to talk about, failed attempts to remember an specific event, playing).

Using the CLAN programs FREQ[4] and KWAL, available through CHILDES, all instances of diminutized words in the transcripts were identified and coded for (a) landing site of diminutive and (b) usage mode. Frozen diminutives, that is, diminutized forms with a distinct, specific and well-known meaning were excluded from analyses (e.g. ***bocaditos*** [BITE-DIM/MASC/PL], used in Lima to refer to hors d'oeuvres; ***fulbito*** [SOCCER-DIM/MASC], a variety of soccer played with fewer players and special rules). Intercoder reliability was established between the authors (both of whom are Spanish–English bilinguals) on 20% of all transcripts using Cohen's kappa (Cohen, 1960). Kappa values are provided in each section.

Landing site. All instances of diminutives were coded for the word-class to which the diminutives were attached. Cohen's kappa yielded almost perfect

[4] FREQ is a program that provides a list of all the words in specified files. We identified all instances of diminutized words and created an include file containing these words. Using this include file, we ran KWAL, a program that locates words in each transcript. The KWAL output provided us with the page and line of each diminutive used in the conversations. We did not code from these outputs: rather, all words were coded from the transcripts and within the conversational context. For more information on CLAN programs the reader is referred to MacWhinney (2000).

interrater reliability (kappa statistic = 0·90). Each diminutized word received one of the following three mutually exclusive codes:

1. *Nouns*: Words referring to animate or inanimate objects in the world, as well as to abstract concepts. Nouns in Spanish can be easily distinguished because they frequently co-occur with definite or indefinite articles. In cases where the article was not present, for example, in possibly ambiguous one-word utterances (e.g. *guapito* [HANDSOME-DIM/MASC]), words were distinguished as either nouns or adjectives using contextual clues and native speaker intuition.
2. *Adjectives*: Words modifying nouns (e.g. *calientito* [HOT-DIM/MASC], *jovencita* [YOUNG-DIM/FEM], *azulita* [BLUE-DIM/FEM]). Here we also included the word *todo* 'all' in each of its variations. *Todo* is considered by the *Real Academia de la Lengua Española*, 'Royal Academy of the Spanish Language' (Alarcos Llorach, 1999), to be an undefined word that can act both as an adjective and as a noun, as in *Toditos los niños vinieron a la fiesta* ('ALL-DIM/MASC/PL the children came to the party') and *Toditos vinieron a la fiesta* ('ALL-DIM/MASC came to the party'). Although *todo* in the latter sentence serves as a noun and occupies the subject position, we maintain that it is in fact modifying an implied noun such as *los niños* 'the children', and thus is best classified as an adjective in both cases.
3. *Adverb*: Words modifying adjectives or verbs (e.g. *ayercito* [YESTERDAY-DIM/MASC], *afuerita* [OUTSIDE-DIM/FEM]).

Usage mode. In addition, each instance of a diminutive was coded for the manner in which it was used within the immediate speech event (i.e. the segmented narrative exchange or the segmented spontaneous conversation exchange), and the speaker (mother or child). Cohen's kappa yielded almost perfect interrater reliability (kappa statistic = 0·93). The following six mutually exclusive codes were used:

1. *Introduced by mother or child*: The speaker used a diminutive spontaneously and for the first time within the speech event. If the speaker introduced a diminutive after having used a nondiminutized form of the same word previously, the second instance was also coded as an introduction.
2. *Self-repetition by mother or child*: The speaker used a diminutive previously introduced and used by him/herself within the same speech event.
3. *Imitation by mother or child*: The speaker used a diminutive form previously introduced by the interlocutor within the same speech event. Here the form had to be an exact copy of the original form (e.g. if the mother uses *todito* [ALL-DIM/MASC], the child must use *todito*

[ALL-DIM/MASC], not *todita* [ALL-DIM/FEM]. In this particular case, *todita* [ALL-DIM/FEM] would be coded as an introduction).

Data analyses

As the total number of diminutives is largely dependent on the total amount of talk, namely on the number of words used during mother–child conversations (i.e. the greater number of words used, the greater the opportunity to diminutize words), percentages rather than frequencies were used as the primary unit of analysis. However, as the majority of words in Spanish cannot be diminutized, we conducted analyses on the mean percentage of diminutives over the total number of diminutizable words produced by each speaker. To determine the total number of diminutizable word tokens, a list of all words produced by mothers and children was obtained. After reviewing the list, all words not diminutizable in Spanish, and particularly in the middle-class Limeño Spanish variety under investigation here, were deleted. These words included all verbs, articles, and some pronouns. In cases where it was not possible to discern the word-class in isolation (e.g. in Spanish, past participles can be used as adjectives or as verbs), we reviewed each word in its sentential and conversational context to determine its status. We were conservative in deleting words from our final list of diminutizable word tokens, asking three speakers of the same middle-class Limeño variety to inform us whether they believed these items to be diminutizable within their variety of Spanish. In cases where there were disagreements between our informants, words were kept in our final list of diminutizable words.

RESULTS

Descriptive overview of diminutive use by dyads

A grand total of 1515 diminutized words were used by dyads, with a mean of 47 diminutives (S.D. = 32·24, range 4–144) per conversation. The mean frequency of diminutives roughly equalled 2% of all words produced in the conversations and 7% of all words allowing diminutive affixes in Limeño Spanish. Of the three different word-classes, the most frequently diminutized were nouns, comprising 80% of all diminutized words, followed by adjectives with 15%, and then by adverbs with 5%.

Within Ambadiang's (1997) three-type classification of Spanish diminutives, 96% of the diminutized words were formed with the *-it-* infix, 3·5% were formed with *-cit-*, and 0·5% were formed with *-ecit-*. Overall, there was consistency in the diminutive forms used, with only three exceptions: *puert-ecita* and *puert-ita* (DOOR-DIM/FEM); *mam-ita* and *mama-cita* (MOTHER-DIM/FEM); *Cristobal-ito* and *Cristobal-cito* (CRISTOBAL-DIM/MASC), with these two last forms produced by the same speaker. All of the forms are acceptable in middle-class, Limeño Spanish. There were

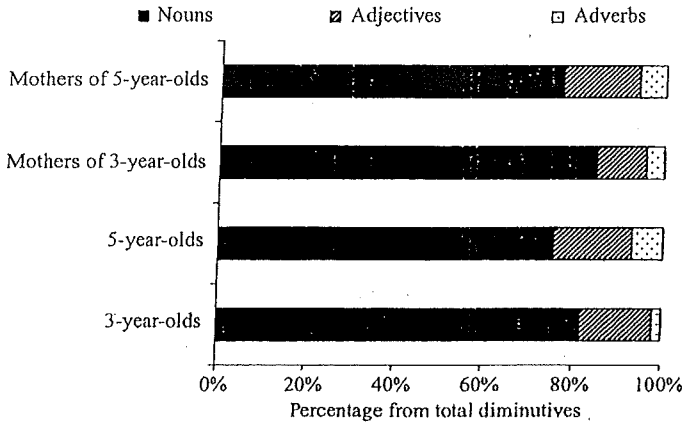


Fig. 1. Percentages for mothers' and children's landing sites of diminutives.

only two errors found in children's diminutized forms. A five-year-old boy used *flor-itas* instead of the more standard adult form, *flor-ecitas* (FLOWER-DIM/FEM/PL). Another five-year-old boy, describing how his friends made fun of a not-so-tall classmate, diminutized *Blanca Nieves* 'Snow White'. In Limeño Spanish, the word would usually not be diminutized.

Child: *el Pepito le dice, 'la familia enano.'*
Pepito says, 'the dwarf family.'
Mother: *¡Oh, pobrecito!*
Oh, poor guy!
Child: *Y dice que ... que es Blanca Nieves, el papá de Blanca Nieves, y el hijito Blancaniecito*
And he says that ... that he is Snow White, the father of Snow White, and the son [is] Little Snow White (SNOW+WHITE+DIM/MASC).
Mother: *¡Uy, qué loco!*
Oh, how crazy!

Despite these two errors, it is of interest that even in the younger age group, there were no overregularizations or other developmental errors in using the diminutive affixes. Finally, in terms of the usage mode of diminutives, the greatest percentage of diminutives were spontaneously introduced (67%), followed by self-repetitions of the same form (19%), and imitation of the interlocutor's form (14%).

Figures 1 and 2 present the distribution of landing site and usage modes for mothers and children by age groups. Two separate multivariate general



Fig. 2. Percentages for mothers' and children's usage mode of diminutives.

linear designs analyses were conducted on these data to compare children's and mothers' use, with landing sites and usage modes as dependent variables and speaker as an independent variable (three-year-olds, five-year-olds, and mothers of three-year-olds and mothers of five-year-olds). To allow comparison, analyses were conducted on mean percentages from total number of diminutives produced by speakers. No significant differences were found for landing site among any of the groups, that is three- and five-year-olds diminutized words in similar ways as did their mothers. A significant multivariate effect for speaker in usage modes was found, $F(6, 118) = 2{\cdot}43$, $p < 0{\cdot}05$. Univariate results revealed a significant effect of speaker on imitation, $F(3, 60) = 3{\cdot}27$, $p < 0{\cdot}05$. *Post hoc* comparisons of the four groups revealed that five-year-olds ($M = 6{\cdot}03$, S.D. $= 8{\cdot}32$) imitated less than the other three groups (for three-year-olds, $M = 11{\cdot}20$, S.D. $= 13{\cdot}65$; for mothers of three-year-olds, $M = 18{\cdot}12$, S.D. $= 17{\cdot}29$; for mothers of five-year-olds, $M = 19{\cdot}34$, S.D. $= 14{\cdot}29$). However, the only difference to reach significance was that between five-year-olds and their mothers.

Age and gender comparisons

This section presents results of analyses of variance and multivariate analyses of variance comparing diminutive use by age and gender of child. To determine whether analyses on mean percentages of diminutives (instead of total frequencies) were warranted, ANOVAs were conducted on dyads' total number of words and of diminutizable words with age group and gender as

TABLE 2. *Dyads' total number of words and total number of diminutizable words by group and gender*

	Three-year-olds ($N=16$) M (S.D.)	Five-year-olds ($N=16$) M (S.D.)
Mother–son		
Total words	3480·13 (1617·95)	3522·75 (1275·50)
Diminutizable words	803·00 (422·57)	780·75 (259·30)
Mother–daughter		
Total words	2189·00 (1141·05)	4106·38 (1061·37)**
Diminutizable words	445·00 (231·74)	910·50 (204·63)***
Totals		
Total words	2834·56 (1507·90)	3814·56 (1172·92)*
Diminutizable words	624·00 (377·58)	845·63 (235·39)*

* $p<0·05$, ** $p<0·01$, *** $p<0·001$.

independent variables. Results showed a significant main effect for group on both total number of words, $F(1, 28)=4·61$, $p<0·05$, and on total number of diminutizable words $F(1, 28)=4·60$, $p<0·05$. Not surprisingly, five-year-olds and their mothers used more words and more diminutizable words in their conversations than three-year-olds and their mothers. Moreover, a significant group × gender interaction effect was found for total number of diminutizable words, $F(1, 28)=5·58$, $p<0·05$. The significant interaction effect was mostly a result of reverse patterns in the number of diminutizable words produced by mother–son and mother–daughter dyads in the two age groups. Among boys, the number of diminutizable words decreased with age, whereas among girls, the number of diminutizable words used increased with age. However, *post hoc* t-test analyses showed significant age-group differences only among mother–daughter dyads in both total numbers of words and of diminutizable words (see Table 2). As significant differences in diminutizable words were found, subsequent analyses were conducted on mean percentages of diminutives; these analyses involved separate comparisons of mothers' and children's use.

Frequency of diminutives. Two separate analyses of variance with age group and gender as independent variables were conducted on mean percentages of total diminutives used by mothers and children. Results showed no significant age-group or gender differences in children's use. A significant main effect for age group was found on mothers' use, $F(1, 28)=8·30$, $p<0·01$. Mothers of three-year-olds used a greater percentage of diminutives ($M=8·80$, S.D. $=3·75$) than mothers of five-year-olds ($M=5·54$, S.D. $=2·35$), regardless of children's gender (see Figure 3).

Landing site. Multivariate general linear designs were used to determine age and gender effects on landing site (i.e. nouns, adverbs, and adjectives).

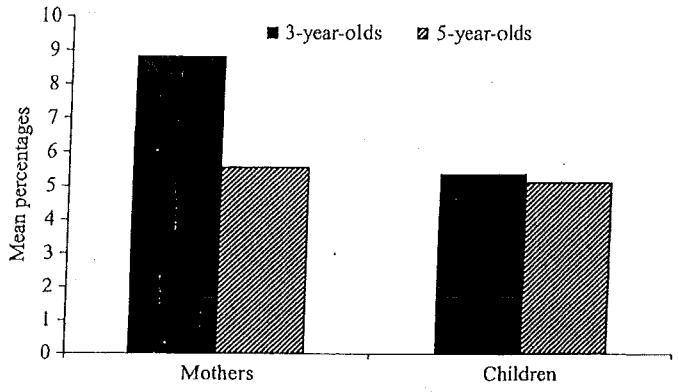


Fig. 3. Mean percentages for total diminutive use by speaker and age group.

Two separate MANOVAs were conducted on children's and mothers' mean percentages of nouns, adverbs and adjectives from total number of diminutizable words. No significant age-group or gender differences were found in children's diminutive landing sites. A significant multivariate effect for age group was found in mothers' landing site of diminutives, $F(3, 26) = 3{\cdot}21$, $p < 0{\cdot}05$. Univariate results revealed a significant effect of age group for nouns, $F(1, 28) = 9{\cdot}49$, $p < 0{\cdot}01$. Mothers of three-year-olds used a greater percentage of diminutized nouns than did mothers of five-year-olds (see Table 3). No significant gender differences were found for mothers' landing sites.

As age-group differences were found in mothers' frequencies of diminutive use, we also conducted analyses on relative frequencies in order to discern possible gender and age effects on diminutives' landing sites while controlling for differences in total number of diminutives produced by mothers. Therefore, an additional MANOVA was conducted using mothers' relative frequencies of landing sites (i.e. type of word from total number of diminutives used) as dependent variables. Once differences in total frequencies were controlled for, the multivariate main effect for age group did not reach significance. However, it is noteworthy that the univariate main effect of age group was still significant for nouns, $F(1, 28) = 4{\cdot}39$, $p < 0{\cdot}05$ (see Table 3).

Usage mode. Multivariate general linear designs were also used to determine age and gender effects on usage mode (introduction, self-repetition, and imitation). Two separate MANOVAs were conducted on children's and mothers' mean percentages of introductions, repetitions and imitations from total number of diminutizable words. No significant age-group or gender differences were obtained in children's usage patterns. A significant multivariate trend for age group was obtained in mothers' usage mode, $F(3, 26) = 2{\cdot}59$, $p < 0{\cdot}10$. Univariate results revealed a significant effect of age group for

TABLE 3. *Mothers' mean percentages of diminutives by landing site and age group*

	Three-year-olds ($N=16$) M (S.D.)	Five-year-olds ($N=16$) M (S.D.)
From total diminutizable words		
Nouns	7·42 (3·32)	4·32 (2·15)**
Adjectives	0·82 (0·74)	0·78 (0·77)
Adverbs	0·56 (0·43)	0·45 (0·37)
From total diminutives used		
Nouns	84·85 (8·17)	76·85 (13·40)*
Adjectives	11·17 (7·57)	17·23 (11·30)
Adverbs	3·98 (4·09)	5·92 (10·35)

* $p<0·05$, ** $p<0·01$.

TABLE 4. *Mothers' mean percentages of diminutives by usage mode and age group*

	Three-year-olds ($N=16$) M (S.D.)	Five-year-olds ($N=16$) M (S.D.)
From total diminutizable words		
Introduction	5·67 (2·71)	3·64 (1·51)*
Self-repetition	1·75 (1·22)	0·86 (0·93)*
Imitation	1·38 (0·89)	1·04 (0·89)
From total diminutives used		
Introduction	64·31 (14·87)	68·50 (17·95)
Self-repetition	17·57 (10·76)	12·17 (10·19)
Imitation	18·12 (17·29)	19·34 (14·29)

* $p<0·05$.

introductions by mothers, $F(1, 28)=6·38$, $p<0·05$, and for mothers' repeated use of diminutized forms, $F(1, 28)=5·10$, $p<0·05$. In both cases, mothers of three-year-olds introduced new forms and repeated their own forms more than mothers of five-year-olds (see Table 4). No significant age group differences were found for imitation, and overall there were no significant gender effects.

As with our previous analyses, in order to discern possible gender and age effects while controlling for differences in total number of diminutives produced by mothers, we also conducted an additional MANOVA on relative frequencies of mothers' usage mode codes (i.e. type of mode from total number of diminutives used). Once differences in total frequencies of diminutive used were controlled for, no age-group (or gender) differences were obtained (see Table 4).

TABLE 5. *Partial correlations between categories for usage mode and total diminutive use*

Variables	1	2	3	4	5	6	7	8
1. Children's total use	—	0·94****	0·85****	0·46**	0·33	0·06	0·28	0·69****
2. Children's introductions		—	0·68****	0·26	0·26	0·03	0·19	0·62***
3. Children's self-repetitions			—	0·32	0·18	−0·10	0·15	0·71****
4. Children's imitation				—	0·52**	0·44*	0·45**	0·19
5. Mothers' total use					—	0·91****	0·84****	0·21
6. Mothers' introductions						—	0·68****	−0·13
7. Mothers' self-repetitions							—	0·04
8. Mothers' imitation								—

* $p<0·05$, ** $p\leqslant 0·01$, *** $p\leqslant 0·001$, **** $p\leqslant 0·0001$.

Relationships between mothers' and children's use of diminutives

To understand the relationship between mothers' and children's use of diminutives, we conducted partial correlation analyses on mean percentages of total use and categories for usage mode (introduction, self-repetition, imitation), controlling for children's age. Children's total use was positively related to mothers' imitation ($r(32)=0·69$, $p\leqslant 0·0001$); that is, the greater the percentage of diminutives the child used, the more the mother imitated her child. Conversely, the more the mother mimicked her child, the more diminutives the child used. A similar relationship was found for mothers' total use of diminutives and child's imitation ($r(32)=0·52$, $p<0·01$) (see Table 5).

To determine the degree to which imitation accounted for total diminutive use in both mothers' and children's language, two separate multiple regression analyses were conducted. Interlocutors' usage mode codes (introduction, self-repetition and imitation) were used as predictors for speakers' total use of diminutives. In both analyses, a dummy variable for age group was entered to control for its effects on total use, and the variables 'introductions' and 'self-repetitions' were combined as a 'spontaneous usage' variable. Results showed a significant R for the regression model on children's total use of diminutives, $F(3, 28)=10·29$, $p\leqslant 0·0001$. Although the three variables combined (i.e. age group, spontaneous usage, and imitations) accounted for 52% ($R^2=0·52$, $Adj\ R^2=0·47$) of the variability in children's total diminutives, only maternal imitation ($sr^2=0·50$) was found to be significant. Results of the regression analyses on mothers' total use of diminutives

were also significant, $F(3, 28) = 8{\cdot}15$, $p < 0{\cdot}001$. The three independent variables combined accounted for 47% ($R^2 = 0{\cdot}47$, $Adj\ R^2 = 0{\cdot}41$) of the variability in mothers' total diminutives; however, only age group ($sr^2 = 0{\cdot}10$) and the variable imitation ($sr^2 = 0{\cdot}18$) were significant. These results provide further evidence that children and mothers encourage the use of diminutives by imitating each other's use.

We also investigated whether the effects of imitation on total use differed by age group (i.e. whether mothers' imitation of children's diminutives had a similar effect on three-year-olds' and five-year-olds' total diminutive use, and whether three-year-olds' and five-year-olds' imitations of mothers' diminutives had a similar effect on mothers' total diminutive use). To address these two questions, an interactive dummy variable was created (consisting of age group with imitation) and then added to the two regression models previously conducted. In both regression models, the coefficient for the interactive dummy variable was not significant, indicating that there were no age-group differences in the effects that imitation had on the interlocutors' total diminutive use.

DISCUSSION

As no previous research has explored the use of diminutives in Spanish-speaking mother–child dyads, the first aim of the study was to describe the basic characterisitics of diminutive use among a group of Spanish-speaking mothers and their preschool children. Our results showed that all dyads spontaneously used diminutives in their conversations, a finding which provides further evidence of the saliency of this complex grammatical form in everyday conversations with children. Overall we found that 2% of all words used and 7% of all diminutizable words were diminutized. This number likely represents the high-end of diminutive use among speakers of Limeño Spanish as the conversational context in our study is characterized by many of the features which are associated with elevated use of diminutive forms, namely female speech, caregiver speech, and child-centred speech events (Ferguson, 1977; Hummel, 1997; Andrews, 1999). Our figure of 2% is notably lower than the 5% reported by Hererra & Carvallo (1987), the only other study which examined Spanish (Venzuelan) diminutive use among children. Although a direct comparison is not possible due to methodological disparities, the differences between our findings and those of Herrera & Carvallo's might point to variations between communities of speakers from diverse Spanish varieties as well as across speech situations. Similarly, our results also support previous work (Andrews, 1999) suggesting that the frequency with which diminutive forms are produced by mothers, and thus heard by children, varies from dyad to dyad. Therefore, although diminutives seem to be used with all children, the saliency of this complex

grammatical morpheme in children's everyday conversational contexts varies along many factors, including the caregiver's particular discourse patterns and the norms of the child's speech community.

Our findings from the examination of diminutive landing sites provide further evidence which supports the existence of universal hierarchies. Recall that these hierarchies suggest, as an example, that if a language allows diminutized adverbs, then it must also allow diminutized items further up on the hierarchy, namely nouns and adjectives; furthermore the hierarchy suggests that items which are higher on the hierarchy will be relatively more productive. Our findings confirm this aspect of the theory, indicating that items sequentially lower on the hierarchy, such as adjectives and adverbs, are far less represented in the sample than nouns, which sit at the top of the cross-linguistic hierarchy. This distribution of landing site was found to be stable across the four speaker groups.

Our findings derived from the examination of usage mode, in turn, indicate that both mothers and children were most likely to introduce diminutives rather than imitate or self-repeat diminutive forms. However, in contrast to our results concerning landing site, we did find a significant difference in the patterns of usage across speakers: once total number of diminutives produced was controlled for, five-year-olds exhibited significantly lower rates of imitation of diminutive forms relative to their mothers. In contrast, no significant differences in the use of imitation were found between three-year-olds' and mothers of either age group. These results seem to suggest that while mothers of five-year-olds continue to imitate and scaffold their children's use of diminutives, five-year-olds are gaining conversational independence, thus exhibiting a weaker tendency to imitate. This finding corresponds to previous work (e.g. Snow, 1986) which suggests that children use imitation as a strategy for expanding their communicative competence; however as children grow older, they achieve greater conversational competence and therefore rely less on imitation relative to other usage forms. However, it is important to note that like previous studies of imitation (Sokolov & Moreton, 1994), in our data we found substantial individual variation in the extent to which imitation was used; this variation likely explains the lack of significant differences between three-year-olds and five-year-olds.

Our second aim was to explore gender and age-group variations in mothers' and children's use of diminutives. Given the strong association between female speech and diminutive forms (Berko Gleason *et al.*, 1994), we had expected to see some gender differences concerning either the speech directed at children or the speech produced by them. However, no significant gender differences were found in any aspect of the data. Nevertheless, gender-related differences in diminutive use do seem to exist in the speech of adult Peruvians; for instance, Caravedo's (1989) corpus of adult conversations

indicates that women make greater use of diminutives than men. Given that no gender differences were found in our data, it is possible that such differences do not become apparent until a later developmental stage (see Melzi & Fernández, 2001), or perhaps are only evident in other speech events and speech situations (e.g. at play among peers, in conversations with fathers) than those under investigation here.

Our findings confirm previous work which suggests that young children as young as three use diminutives in similar ways as their adult interlocutors (Rūķe-Draviņa, 1976; Gillis, 1997; Stephany, 1997). We found that when the total amount of speech was controlled for, three-year-old children used diminutives with similar frequencies and distributions as five-year-old children. Of 557 diminutive formations produced by children in the present study, there were only two instances of use which were ill-formed or which differed from adult use. Furthermore, while children used diminutive morphology with close to no errors, they did seem to make occasional mistakes with other aspects of grammatical morphology, such as verb tense. It should be noted, however, that pragmatic or semantic use of the diminutive was outside of the scope of the present study; it might be the case that had these aspects been considered, developmental differences would have emerged.

While no developmental differences on the part of the child were apparent here, significant variation was found in mothers' use of diminutized forms across age groups. Overall we observed that mothers of three-year-old children used more diminutives than mothers of five-year-old children. In particular, mothers of three-year-olds also introduced more diminutives into the conversation and repeated their own diminutive forms more often than did mothers of five-year-olds. This latter finding, however, is largely a reflection of the fact that mothers of three-year-olds simply produced more diminutives than mothers of five-year-olds. This finding fits with previously noted arguments that diminutives signal a child-centred, nurturing and intimate environment. Perhaps in part due to this association, Ferguson (1977) identified diminutives as a systematic feature of child-directed speech (CDS or babytalk register). His work in this area suggests that mothers' use the babytalk register from children's first months of life until about their fourth birthday, although there is substantial variation across cultures, speakers, and contexts. Results of our study provide evidence that diminutives are, in fact, an aspect of Peruvian Spanish child-directed speech and more importantly, that their use decreases during the preschool years as children gain conversational and linguistic competence. Similar results have been provided by Berko Gleason *et al.* (1994) who showed that English-speaking parents decreased their use of diminutized forms as children approached three years of age. The age differences between Berko Gleason *et al.*'s (1994) and our results potentially reflect cross-linguistic variation in the extent to which diminutives are a salient aspect of CDS.

The third aim of the present study was to investigate the relationship between mothers' and children's patterns of diminutive use. Our findings here clearly indicate that certain diminutive usage modes are more critical than others in terms of promoting overall diminutive use. More precisely, the frequency of mother-to-child imitations of diminutive forms was a clear predictor of total use of diminutized forms by the child. Conversely, mothers' total use of diminutives was predicted by children's imitation and by children's age. It is important to note that although five-year-olds tended to use imitation less frequently than their mothers, there were no differences between the groups in the role imitation played for their interlocutors' overall use of diminutives.

The strong relationship between diminutive imitation and overall use highlights the sensitivity of both interlocutors to each other's speech. As the mother picks up her child's diminutives and imitates them back to him or her, she appears to reinforce the child's use of diminutized forms and thereby to facilitate greater use of diminutives. While such imitation appears to play a powerful role in accounting for total frequency of diminutives used, it is not clear what processes underlie this relationship. It seems likely, however, that the imitation of diminutives by interlocutors serves to establish – or possibly reflect – the intimate, playful, warm, or loving nature of the interaction, which ultimately results in greater total use of diminutive forms.

This finding is reminiscent of the connection between attachment and language development posited by some scholars of child language (Locke, 1995; Nelson, 2001). Locke, for instance, argues that attachment between children and their caregivers is both the impetus and the foundation for children's development of language. One of the hallmarks of the close bond between infant and caregiver is the increased attention caregivers and children pay to each other, in particular to each other's speech. This attention, which is manifested in caregivers' contingent responses to and imitations of children's verbalizations and actions, increases infants' sensitivity to the caregivers' speech. In this respect, Pawlby (1977, cited in Locke, 1995) notes that '... when mothers themselves imitate an action which the child has just performed, (t)he infant's action is ... 'highlighted' or 'marked out' as something special' (p. 220), leading to the infants' increased production of the same action. A similar process seems to be at play with evaluative morphology: the more the mother imitates her child's diminutized forms, the greater the child's use of them. Since the children in this study are older than those included in previous investigations concerning imitation, our results seem to suggest that mother-to-child imitation continues to play a significant role in children's language development for much longer than previously considered. In addition, the finding that child-to-mother imitations are also highly correlated to mothers' total use of diminutives speaks to the fact that

children are active agents in the co-creation of a nurturing environment which might promote their development of language.

CONCLUSION

The present study sheds new light on how mothers and children use one aspect of evaluative morphology in conversations with each other, and to what extent these usage patterns vary by age and gender. We find that by three years of age, children use diminutives in similar ways as both the five-year-olds and the mothers in our study. While there were no significant gender differences, there were significant variations across age groups, with mothers of three-year-olds using a greater overall frequency of diminutives than mothers of five-year-olds. These results show that diminutives are a salient feature of Peruvian mother–child speech, which decreases between the ages of three and five. Furthermore, we find that imitation of diminutive forms by one interlocutor is highly correlated and serves as a strong predictor of total frequency of diminutive use by their interlocutor.

The present study suggests several areas for future research. Perhaps most saliently, this paper underlines the need to take careful consideration of both the micro and macrocontext of language use. Here, for instance, each diminutive is analysed within the microcontext of a specific narrative exchange, as well as within the wider, macrocontext of an extended mother–child conversation. Use of these frames in coding and analysis allowed us to uncover relationships, such as that between imitation of one interlocutor and total frequency of use by the other, which otherwise would not have been visible. This relationship was also made apparent through the inclusion of a relatively large number of participants in our analyses.

Given that three-year-olds used diminutives in roughly the same ways as five-year-olds and their mothers, it would also be of interest to collect and analyse similar data among younger children in order to investigate the developmental trajectory of both the landing site of the diminutive affix and the mode of usage prior to three years of age. Furthermore, as noted above, pragmatic and semantic usage patterns were outside the scope of this study. As some of the complexity of diminutive use lies in these two domains, it would be of interest to analyse how three- and five-year-olds potentially differ in their semantic and pragmatic manipulation of this form.

Finally, as diminutives seem to play an important role in marking speech events as child-centred, intimate, and supportive, as well as in introducing children into the wider, more complex arena of grammatical morphology, it is surprising that fewer empirically based studies of evaluative morphology exist. This imbalance likely corresponds to the dominance of English – a relatively impoverished language in terms of evaluative morphology – as the

primary language for investigations of child language development. Our study thus provides further support for the recent calls (e.g. Lieven, 1994) for increased study of non-English languages in the field.

REFERENCES

Alarcos Llorach, E. (1999). *Gramática de la lengua española*. Madrid: Real Academia Española.

Alonso, A. (1961). Concept, emotion, action, and fantasy in diminutives. *Estudios lingüísticos: temas españoles*. Madrid: Gredos.

Ambadiang, T. (1997). Las bases morfológicas de la formación de diminutivos en español. *Verba* **24**, 99–132.

Andrews, E. (1999). Gender roles and perception: Russian diminutives in discourse. In M. Mills (ed.), *Slavic gender linguistics*. Philadelphia: John Benjamins.

Bates, E. & Rankin, S. (1979). Morphological development in Italian: connotation and denotation. *Journal of Child Language* **6**, 29–52.

Bauer, L. (1997). Evaluative morphology: in search of universals. *Studies in Language* **21**, 533–75.

Berko, J. (1958). The child's learning of English morphology. *Word* **14**, 150–77.

Berko Gleason, J., Perlmann, R. U., Ely, R. & Evans, D. W. (1994). The babytalk register: parents' use of diminutives. In J. L. Sokolov & C. A. Snow (eds), *Handbook of research in language development using CHILDES*. Hillsdale, NJ: Erlbaum.

Bulatetska, L. (1996). *Linguistic means of expressing distance between interlocutors in Ukrainian* (ERIC Document Reproduction Service No. ED 394 346).

Caravedo, R. (1989). *El español de Lima: materiales para el estudio del habla culta*. Lima: Fondo Editorial de la Pontificia Universidad Católica.

Ceccherini, M., Bonifacio, S. & Zocconi, E. (1997). Acquisition of diminutives in Italian. In W. U. Dressler (ed.), *Studies in pre- and protomorphology*. Vienna: Verlag.

Cohen, J. (1960). A coefficient of agreement for nominal scales. *Educational and Psychological Measurement* **20**, 37–46.

Daltas, P. (1985). Some patterns of variability in the use of diminutive and augmentative suffixes in spoken Modern Greek Koine (MGK). *Glossologia* **4**, 63–88.

Derwing, B. L. (1976). Morpheme recognition and the learning of rules for derivational morphology. *Canadian Journal of Linguistics* **21**, 38–66.

Dressler, W. U. & Barbaresi, L. M. (1994). *Morphopragmatics: dimimutives and intensifiers in Italian, German, and other languages*. New York: Mouton de Gruyter.

Ettinger, S. (1974). *Diminutiv- und Augmentativbildung: Regeln und Restriktionen*. Tübingen: Narr.

Ferguson, C. A. (1977). Babytalk as a simplified register. In C. E. Snow & C. Ferguson (eds), *Talking to children*. New York: Cambridge University Press.

García Márquez, G. (1967). *Cien años de soledad*. Buenos Aires: Editorial Sudamericano.

Gillis, S. (1997). The acquisition of diminutives in Dutch. In W. U. Dressler (ed.), *Studies in pre- and protomorphology*. Vienna: Verlag.

Herrera, M. & Carvallo de P. Y. (1987). El diminutivo en niños caraqueños de 7 10 años de edad: un enfoque psicolingüístico. *Investigación y Posgrado* **2**, 53–66.

Hummel, M. (1997). Para la lingüística de vuestro diminutivo: los diminutivos como apreciativos. *Anuario de Estudios Filológicos* **XX**, 191–210.

Inchaurralde, C. (1997). Space, reference, and emotional involvement. In S. Niemeier & R. Dirven (eds), *The language of emotions: conceptualization, expression, and theoretical foundation*. Amsterdam: John Benjamins.

Jaeggli, O. (1980). Spanish diminutives. In F. Nuessel (ed.), *Contemporary studies in Romance languages*. Bloomington: Indiana University Linguistics Club.

Jurafsky, D. (1996). Universal tendencies in the semantics of the diminutive. *Language* **72**, 533–78.

Lieven, E. V. M. (1994). Crosslinguistic and crosscultural aspects of language addressed to children. In C. Gallaway & B. J. Richards (eds), *Input and interaction in language acquisition*. Cambridge: CUP.

Lipski, J. M. (1994). *Latin American Spanish*. New York: Longman.

Locke, J. L. (1995). Development of the capacity for spoken language. In P. Fletcher & B. MacWhinney (eds), *The handbook of child language*. Oxford: Blackwell.

MacWhinney, B. (2000). *The CHILDES project: tools for analysing talk* (3rd edn.). Mahwah, NJ: Erlbaum.

Melzi, G. & Fernández, C. (2001, April). Emotion talk in Peruvian mother–child conversations. Poster presented at the Society for Research in Child Development, Minneapolis, MN.

Nelson, K. (2001, April). Levels of representation and consciousness: implications for attachment theory. Paper presented at the Society for Research in Child Development, Minneapolis, MN.

Nieuwenhuis, P. (1985). Diminutives. Unpublished doctoral dissertation, University of Edinburgh, Edinburgh, United Kingdom.

Olmsted, H. (1994). Diminutive morphology of Russian children: a simplified subset of nominal declension in language acquisition. In *Alexander Lipson in Memoriam*. Bloomington: Slavica Publishers.

Pawlby, S. J. (1977). Imitative interaction. In H. R. Schaffer (ed.), *Studies in mother–infant interaction*. New York: Academic Press.

Peters, A. M. (1995). Strategies in the acquisition of syntax. In P. Fletcher & B. MacWhinney (eds), *The handbook of child language*. Oxford: Blackwell.

Rūķe-Draviņa, V. (1976). 'Mama' and 'papa' in child language. *Journal of Child Language* 3, 157–66.

Sifianou, M. (1992). The use of diminutives in expressing politeness: modern Greek vs. English. *Journal of Pragmatics* 17, 155–73.

Snow, C. E., Smith, N. S. H. & Hoefnagel-Höhle, M. (1980). The acquisition of some Dutch morphological rules. *Journal of Child Language* 7, 539–53.

Snow, C. E. (1986). Conversations with children. In P. Fletcher & M. Garman (eds), *Language acquisition*. Cambridge: CUP.

Sokolov, J. L. & Moreton, J. (1994). Individual differences in linguistic imitativeness. In J. L. Sokolov & C. A. Snow (eds), *Handbook of research in language development using CHILDES*. Hillsdale, NJ: Erlbaum.

Stephany, U. (1997). Diminutives in early child Greek, a preliminary investigation. In W. U. Dressler (ed.), *Studies in pre- and protomorphology*. Vienna: Verlag.

Zuluaga, A. O. (1991). La función del diminutivo en español. *Thesaurus* 1, 305–30.